

# Quarto and MakeTables

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This document illustrates how to use MakeTables to write academic papers. Just write your paper in markdown in Jupyter Notebooks. Tables are displayed as HTML interactively when you work in the notebook on the screen. When you render the notebook as PDF with Quarto, MakeTables generates LaTeX or Typst code, depending on the output format.

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# 1 Introduction

A very convenient way to write academic papers which include the code for all reported analyses is to use Jupyter notebooks together with [Quarto](#). Quarto is a scientific publishing system that allows you to combine text, code, and results in a single document. Quarto renders Jupyter notebooks to PDF using either LaTeX or Typst. When you include tables created with MakeTables in your notebook, they will be rendered directly on the screen in the notebook (as HTML via [Great Tables](#)) but will be rendered as LaTeX or Typst tables in the final PDF document by Quarto.

Here is simple example of how to use MakeTables in a notebook to write a paper. It generates tables, for instance based on regressions in either [pyfixest](#) or [statsmodels](#). Render this notebook with `quarto render quartoExample.ipynb --to pdf` to create the PDF document via LaTeX, or with `quarto render quartoExample.ipynb --to typst` to create it via Typst.

Note it is also very easy to include references and citations in your Quarto document (see for instance Wooldridge (2010) and Greene (2018)).

## 2 Some Tables

Table 1 shows some descriptive statistics.

Table 1: Automotive Data Summary

|               | Domestic |          |           | Foreign |          |           |
|---------------|----------|----------|-----------|---------|----------|-----------|
|               | N        | Mean     | Std. Dev. | N       | Mean     | Std. Dev. |
| Mileage (mpg) | 52.00    | 19.83    | 4.74      | 22.00   | 24.77    | 6.61      |
| Weight (lbs.) | 52.00    | 3,317.12 | 695.36    | 22.00   | 2,315.91 | 433.00    |
| Price         | 52.00    | 6,072.42 | 3,097.10  | 22.00   | 6,384.68 | 2,621.92  |

Table 2 shows regression results here estimated with the excellent `pyfixest` package.

Table 2: A Regression Table (PyFixest)

|               | Mileage (mpg)        |                      |
|---------------|----------------------|----------------------|
|               | (1)                  | (2)                  |
| Weight (lbs.) | -0.006***<br>(0.001) | -0.004**<br>(0.002)  |
| Length (in.)  |                      | -0.080<br>(0.055)    |
| Intercept     | 39.440***<br>(1.614) | 47.885***<br>(6.088) |
| Observations  | 74                   | 74                   |
| $R^2$         | 0.652                | 0.661                |

Significance levels: \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Format of coefficient cell: Coefficient (Std. Error)

Table 3 uses `statsmodels` instead to estimate a probit model.

Table 3: A Regression Table (Statsmodels)

|               | Foreign (indicator)  |                      |
|---------------|----------------------|----------------------|
|               | OLS                  | Probit               |
|               | (1)                  | (2)                  |
| Weight (lbs.) | -0.001***<br>(0.000) | -0.004***<br>(0.002) |
| Length (in.)  | 0.007<br>(0.005)     | 0.033<br>(0.045)     |
| Price         | 0.000***<br>(0.000)  | 0.001***<br>(0.000)  |
| Intercept     | 0.656<br>(0.583)     | 1.327<br>(4.992)     |
| Observations  | 74                   | 74                   |
| $R^2$         | 0.552                | -                    |
| Pseudo $R^2$  | -                    | 0.606                |

Significance levels: \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Format of coefficient cell: Coefficient (Std. Error)

### 3 Conclusion

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## 4 References

Greene, W. H. (2018). *Econometric analysis* (8th ed.). Pearson.

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*.